

Homework 3

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Part 1. Set up tasks

```
# reading in packages
library(tidyverse)
library(janitor)
library(here)

# reading in kelp data
kelp <- read.csv(
  here("data", "temp-kelp.csv")
)

# reading in personal data finalized
personal_data <- read.csv(
  here("data", "personal_data.csv")
)
```

Part 2. Problems

Problem 1. Giant kelp fronds

Climate change threatens all ecosystems, including giant kelp (*Macrocystis pyrifera*) forests.

You are interested in the potential relationship between ocean temperature (measured in °C) and giant kelp frond elongation rate (also known as growth rate, measured in cm day⁻¹).

You conduct a field study in which you track giant kelp individuals growing in different temperatures (on average) and measure their elongation rates. Admittedly, this is not a perfect study, but you do the best with what you have!

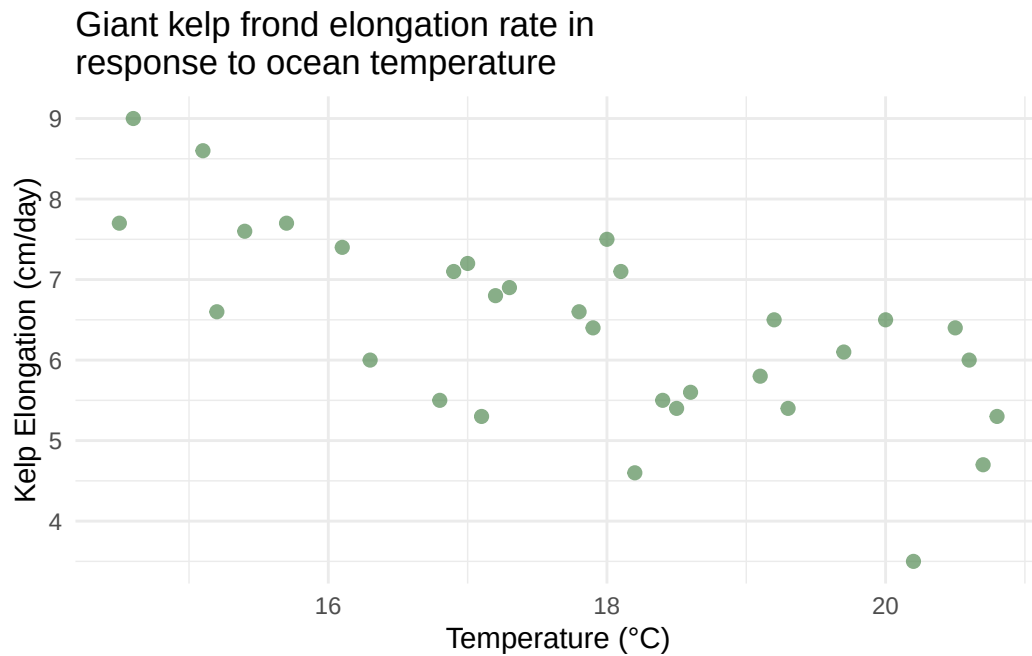
a. An appropriate test

The appropriate tests to determine the strength of the relationship between ocean temperature and giant kelp frond elongation rate are Pearson's correlation (r) and Spearman's rank correlation (ρ) test. Pearson's correlation (r) evaluates a linear relationship between the continuous variables and assumes normality of the data, whereas Spearman's rank correlation (ρ) assesses the monotonic relationship by ranking the data points. Since biological growth rates like kelp elongation often do not follow a typical linear pattern, Spearman's rank correlation (ρ) may provide a more accurate measure of the relationship strength.

b. Create a visualization

```
# base layer: ggplot
ggplot(kelp, # from data source
      mapping = aes(
        x = temp_c, # x axis temp
        y = kelp_elong # y axis elongation
      )) +
# first layer : data points showing individual observation
geom_point(size = 2,
           alpha = 0.7,
           color = "palegreen4") + # adding custom color
# renaming axes, adding a title
labs(
  x = "Temperature (°C)",
  y = "Kelp Elongation (cm/day)",
  title = "Giant kelp frond elongation rate in
response to ocean temperature"
) +
```

```
# adding theme
theme_minimal()
```



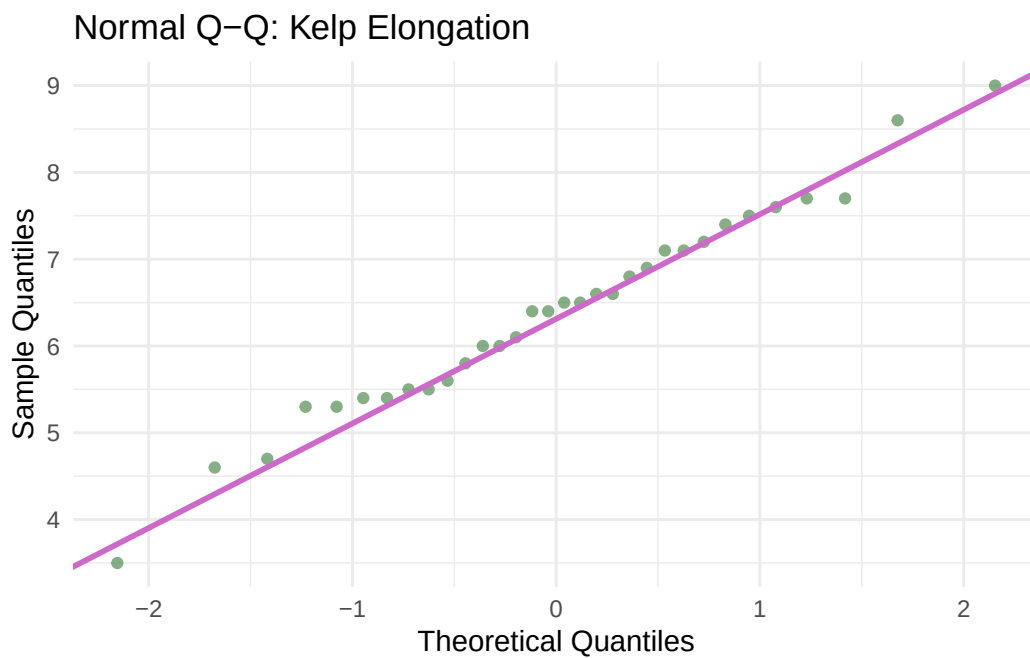
Ocean temperature increase leads to an approximate decrease in kelp elongation.

c. Check your assumptions and run your test

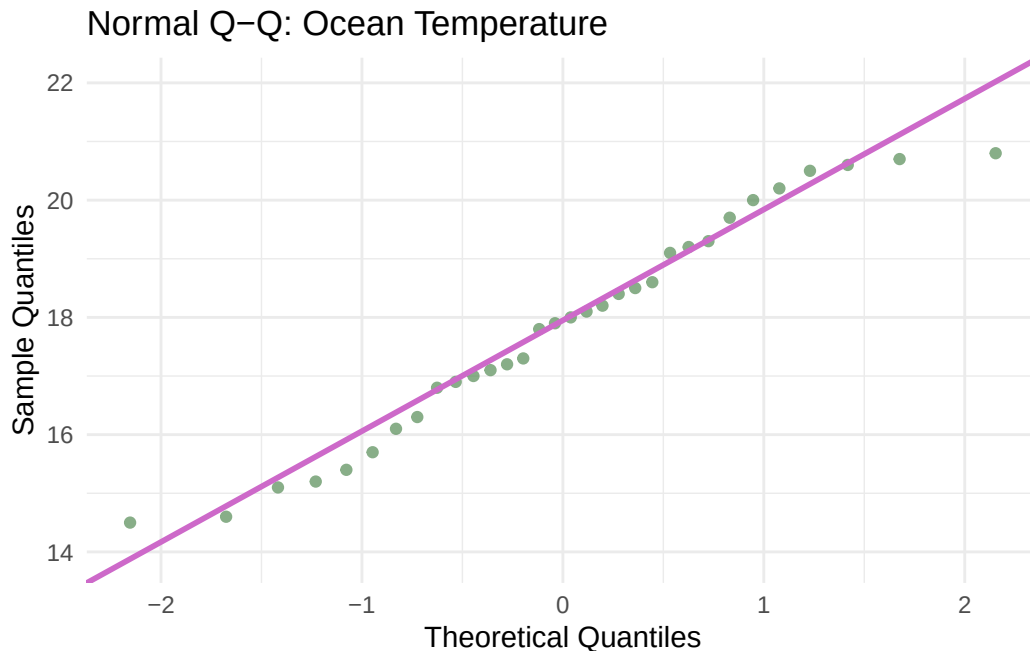
Checking assumptions for Pearson's correlation (r)

```
# qq plot for kelp elongation data
qq_kelp <- ggplot(data = kelp, aes(sample = kelp_elong)) +
  # creating qq points, adding custom color
  stat_qq(color = "palegreen4", alpha = 0.7) +
  # qq_line, adding custom color
  stat_qq_line(color = "orchid3", linewidth = 1) +
  # adding title and axes
  labs(title = "Normal Q-Q: Kelp Elongation",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  # adding a theme
  theme_minimal()
```

```
# qq plot for ocean temperature
qq_temp <- ggplot(data = kelp, aes(sample = temp_c)) +
  # creating qq points, adding custom color
  stat_qq(color = "palegreen4", alpha = 0.7) +
  # qq_line, adding custom color
  stat_qq_line(color = "orchid3", linewidth = 1) +
  # adding title and axes
  labs(title = "Normal Q-Q: Ocean Temperature",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  # adding a theme
  theme_minimal()
# looking at both plots
qq_kelp
```



```
qq_temp
```



Assumptions Assessment

- **Continuous:** Both ocean temperature (measured in °C) and giant kelp frond elongation rate (measured in cm day⁻¹) are numeric, continuous ratio-scale variables. This satisfies the primary data structure requirement for a parametric correlation test.
- **Linearity:** Visual inspection of the initial scatterplot shows a generally straight-line association between temperature and kelp growth rate. There is no clear evidence of severe quadratic or non-linear curvature, satisfying the assumption of linearity.
- **Normality:** Despite some deviance in tails of both QQ-plots, the points generally follow the reference line, allowing for the assumption of normality to be met.
- **Independent observations:** The data points were collected from distinct, individual giant kelp organisms tracked during a field study. Because measurements are not repeated on the same individuals over time or paired structurally, the assumption of independent observations is met.

Since the assumptions of continuous variables, linearity, normality, and independent observations are met, the use of a Pearson's correlation test (r) is appropriate for the data.

Running a Pearson's correlation test (r)

```
# kelp_pearson to show parametric test
kelp_pearson <- cor.test(kelp$temp_c,
                        kelp$kelp_elong,
```

```

# pearson's correlation
method = "pearson")
# viewing correlation test results
kelp_pearson

```

Pearson's product-moment correlation

```

data: kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
      cor
-0.6877489

```

d. Results communication

To evaluate the strength of the relationship between between temperature and giant kelp frond elongation rate I used a Pearson's correlation test (r) due to the model matching assumptions of linearity, normality, continuous variables, and independent observations. We found a moderate relationship between ocean temperature and kelp elongation (Pearson's $r = -0.69$, 95% CI $[-0.84, -0.45]$, $t(30) = -5.19$, $p < 0.001$, $\alpha = 0.05$). This indicates that there is a negative relationship between temperature and kelp elongation where for every 1 °C increase, kelp elongation decreases by 0.43 cm day⁻¹.

e. Test implications

Our analysis confirms a significant, moderate-strong negative relationship between ocean warming and giant kelp productivity, with kelp frond elongation rates decrease by 0.43 cm day⁻¹ for every 1 °C increase. This decline indicates that increasing ocean temperatures are directly damaging giant kelp habitats, which without proper action may result in greater loss of biodiversity as species that rely on giant kelp forests become at risk. To protect this ecosystem, a prioritization of restoration and monitoring of giant kelp habitats experiencing high thermal stress.

f. Double check your own work

Running a Spearman's rank correlation test

```
# kelp_spearman to display rank test
kelp_spearman <- cor.test(kelp$temp_c,
                          kelp$kelp_elong,
                          # spearman's correlation
                          method = "spearman")
# viewing correlation test results
kelp_spearman
```

Spearman's rank correlation rho

```
data: kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.6891684
```

A parametric Pearson's correlation test revealed a moderate negative linear relationship between ocean temperature and kelp elongation rate ($r = -0.69$, 95% CI $[-0.84, -0.45]$, $t(30) = -5.19$, $p < 0.001$, $\alpha = 0.05$). Similarly, a non-parametric Spearman's rank correlation test confirmed a moderate negative monotonic relationship between ocean temperature and kelp elongation rate ($\rho = -0.69$, $S = 9216.1$, $p < 0.001$). Both tests result in a rejection of the null hypothesis ($\alpha = 0.05$), indicating that whether the data are analyzed linearly or monotonically, we arrive at the same conclusion: increasing ocean temperatures correlate with a clear decline in kelp growth.

Problem 2. Personal data

a. Updating visualizations

Figure A: Productivity Score By Location

```
# creating new object from my personal data
personal_clean <- personal_data |>
clean_names() |>
# selecting for desired columns
select(productivity_scale,
location,
```

```

duration_minutes,
breaks_minutes)
# base layer: ggplot
ggplot(personal_clean,
# making x axis location and y axis productivity
  mapping = aes(x = location,
                 y = productivity_scale,
                 # coloring based on location
                 color = location)) +
# first layer: boxplot
geom_boxplot() +
# second layer: jitter
geom_jitter(height = 0.1,
            width = 0.25,
            alpha = 0.7) +
# changing x and y axis name, adding a title
labs(
  x = "Study Location",
  y = "Productivity Score (1 - 4)",
  title = "Location impact on productivity",
  caption = "Dataset last updated: 2026-05-26.") +
# changing colors of location
scale_color_manual(values = c(
  "library " = "palegreen4",
  "home" = "orchid3"
)) +
# adding theme
theme_minimal() +
# removing legend
theme(legend.position = "none")

```

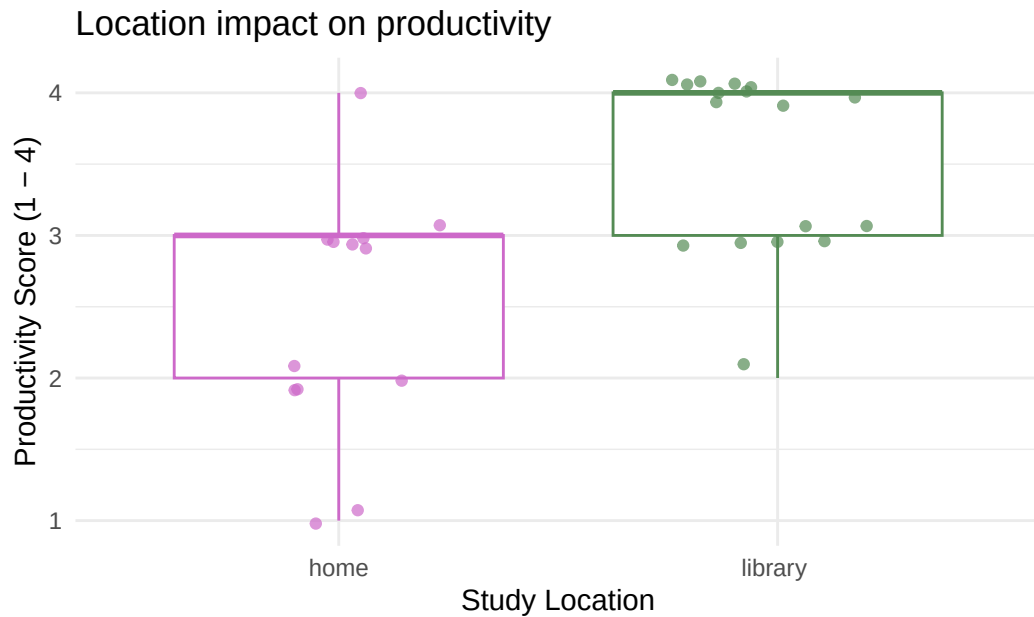



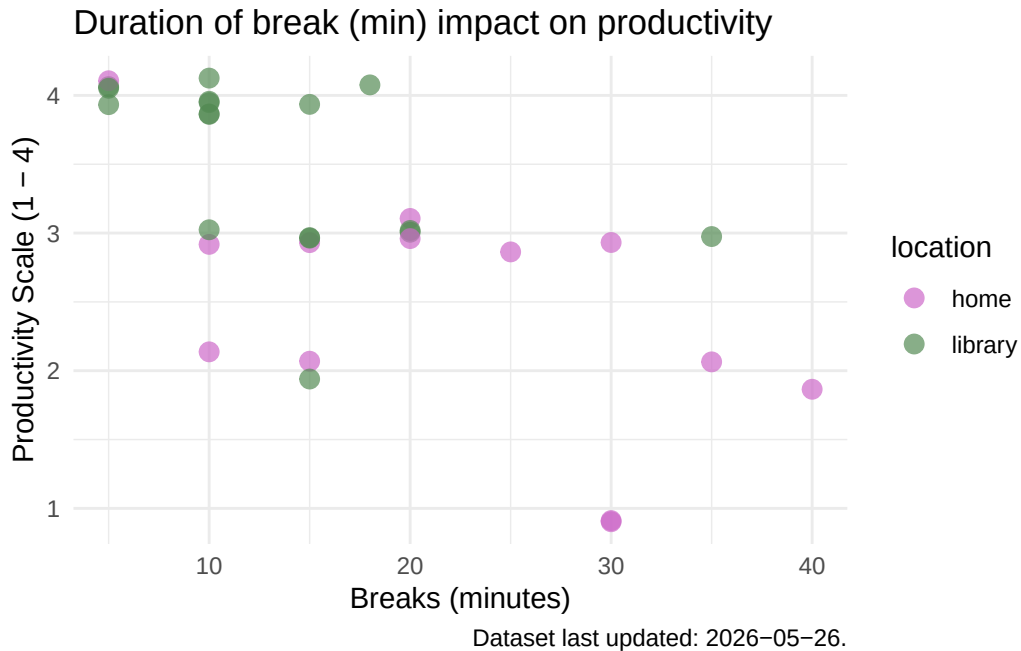
Figure B: Productivity Score By Breaks

```
# base layer: ggplot
ggplot(personal_clean,
# making x axis total duration, y axis productivity scale
# coloring based on location
mapping = aes(x = breaks_minutes,
               y = productivity_scale,
               color = location)) +
# first layer: geom_point
geom_point(size = 3,
           alpha = 0.7,
           # changing position to increase visibility of individual points
           position = position_jitter(width = 0,
                                       height = 0.15)) +
# renaming axis, adding a title
labs(
  x = "Breaks (minutes)",
  y = "Productivity Scale (1 - 4)",
  title = "Duration of break (min) impact on productivity",
  caption = "Dataset last updated: 2026-05-26.") +
# changing colors
scale_color_manual(values = c("home" = "orchid3",
```

```

"library " = "palegreen4")) +
# adding a theme
theme_minimal()

```



b. Captions

Figure A: Productivity scores differ across study environments. Jittered points represent individual observations from self reported study sessions (orchid: home, green: library), with data current through May 26, 2026. Productivity is evaluated on a tiered percentage scale where 1 indicates 50% or less productivity, 2 indicates 50–67%, 3 indicates 67–83%, and 4 indicates 83% or more. Overlaid box plots display a summary of the data, where horizontal bolded lines indicate median productivity score for each location, while the upper and lower box boundaries represent the interquartile range (IQR).

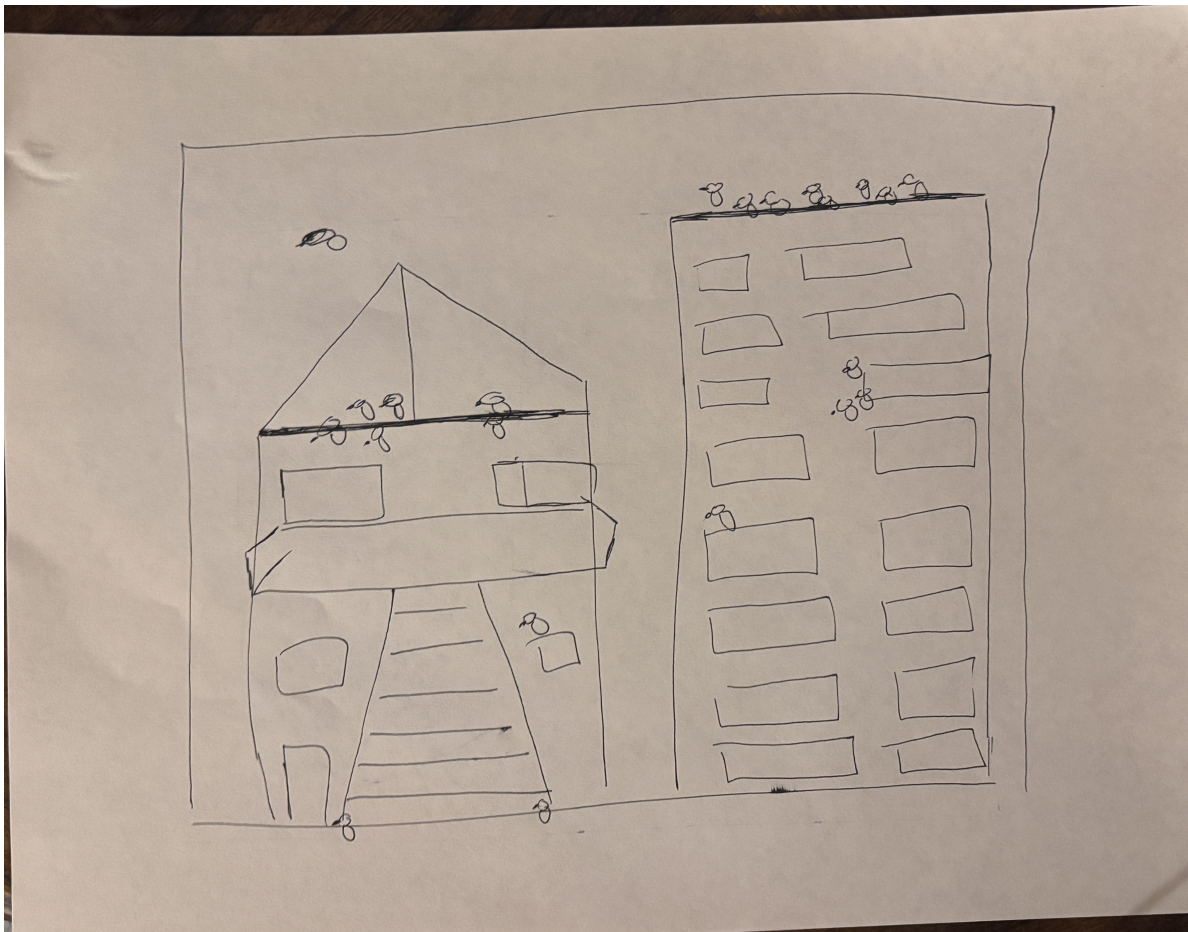
Figure B: Productivity scores differ across varying break durations and study environments Individual points represent distinct recorded study sessions, with data current through May 26, 2026. Productivity is evaluated on a tiered percentage scale where 1 indicates 50% or less productivity, 2 indicates 50–67%, 3 indicates 67–83%, and 4 indicates 83% or more. Point colors correspond to the session location (orchid: home, pale green: library), plotted continuously along the x-axis to show how the duration of breaks taken (in minutes) relates to subsequent productivity ratings (1–4).

Problem 3. Affective Visualization

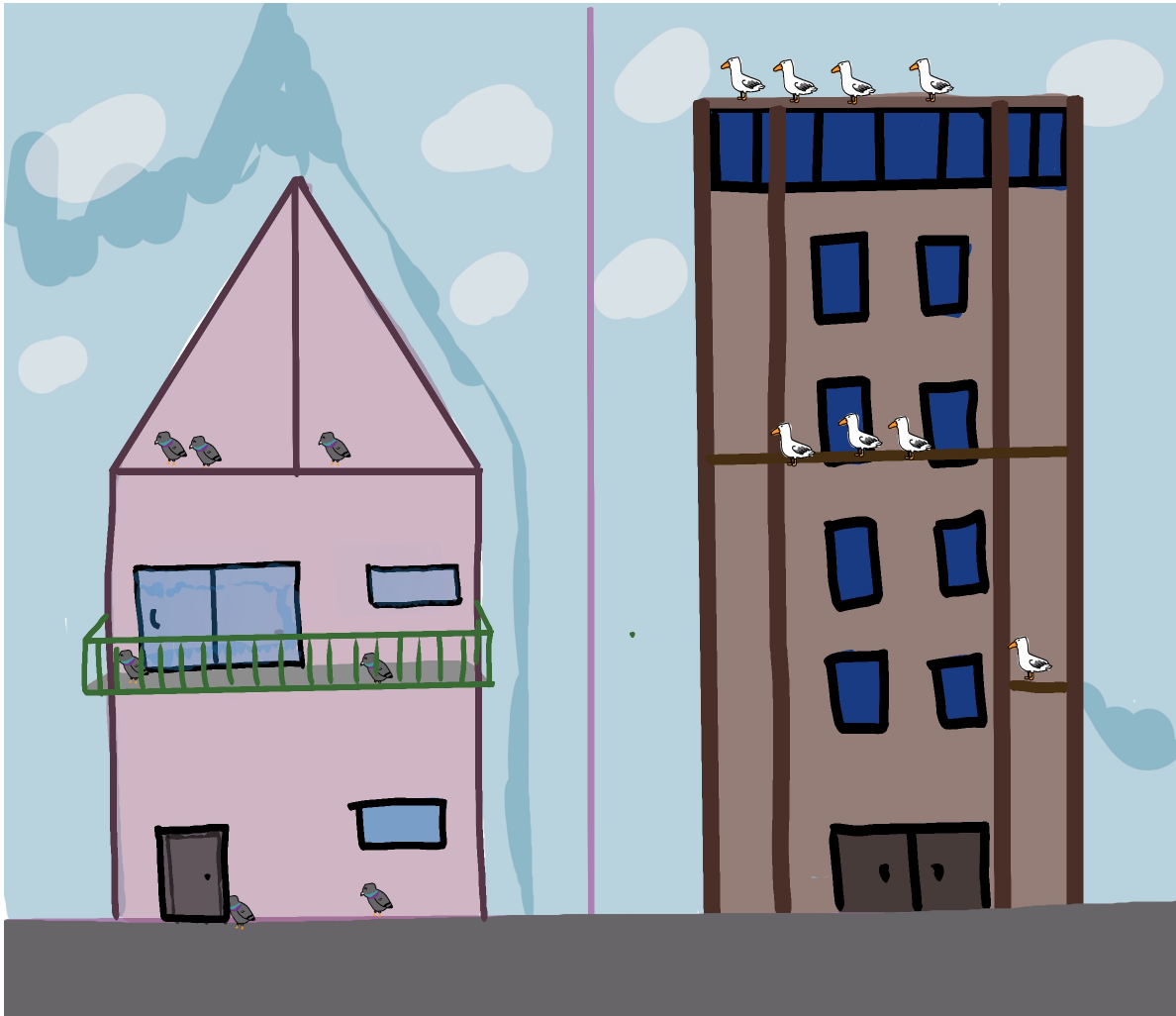
a. Describe in words what an affective visualization could look like for your personal data (3-5 sentences).

An affective visualization for my personal data may look like drawings of my comparison locations within my visualization. Through utilizing the outline of my boxplots generated in Figure A, I can use the IQR line of the home data set as the roof peak, and the median line for both data sets as details of the buildings. For the home drawing I will include details of my current home to indicate the personal nature of this data. For the library drawing I will attempt to make it look similar to the outside of the UCSB Davidson library to again depict how this data correlates to my life. Individual data points will be indicated by pigeons (home) and seagulls (library).

b. Create a sketch (on paper) of your idea.



c. Make a draft of your visualization.



d. Write an artist statement.

The content of my piece displays my home in Isla Vista and the UCSB Davidson library overlaid on the box plots of my data. I then included birds to represent the individual observations. I was inspired by Jill Pelto and how she creates art inspired by what her statistical analysis describes. The medium of my art is a digital drawing, done on the platform Procreate.

e. Prep your materials to share in class.

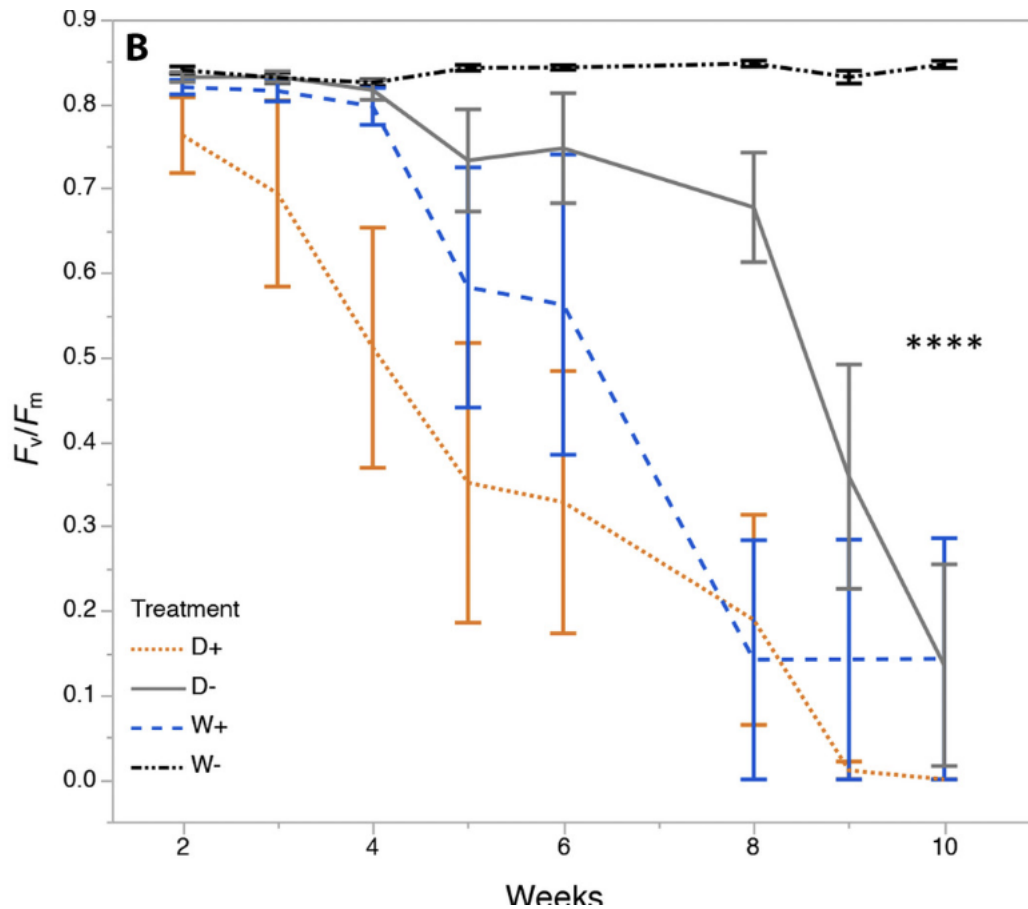
[View slideshow](#)

Problem 4. Statistical critique

a. Revisit and summarize

The authors used a two-way ANOVA test to compare the mean of the florescence (F_v / F_m) based on the treatment given (drought and fungal inoculation, drought without inoculation, watering and inoculation, and watering without inoculation). In running a two-way ANOVA the authors were able to demonstrate that there was a difference in means between the groups through having a p-value < 0.001 . In understanding how fluorescence is impacted by treatment, the authors of this journal are able to address whether increased drought stress is causing decreased resistance to pathogens in *Arctostaphylos glauca*.

Figure 2: B



b. Visual clarity

The figure shown depicts nonlinear trend lines of the florescence based on treatment, differentiated through different color and line types, which are indicated in a legend. The x and y axes

are in a logical position, with time being on the x-axis, allowing the change in florescence as time goes on (in weeks) to be clear to the viewer. Additionally each line depicts $\pm$ SE bars around the florescence values. Underlying data is not shown, which if displayed may have provided more clarity for the viewer to what is influencing the figure lines.

c. Aesthetic clarity

The authors minimized visual clutter by not displaying underlying values, which may have made the figure harder to interpret or read due to four treatment lines being displayed. Despite this, the $\pm$ SE bars add some clutter as bars overlap, making some treatments more difficult to interpret. Regarding the data:ink ratio, the data is slightly more prevalent as the lines and $\pm$ SE bars are the dominant feature of the figure, but the ink is still notable yet kept minimal in the location of the legend being strategically placed in the bottom left of the figure.

d. Recommendations

I would recommend for the authors to change some of the colors of their trend lines to make more visibly appealing. In having three different colors and three different line types shown it makes this visual aspect of the figure unappealing to the viewer. Additionally the authors may consider increasing the height of the figure so that there is less overlap between the SE bars. Also including the points in which the SE bars reference may help the viewers better understand the data. Finally, the authors may consider removing the legend and instead describing which treatment each color corresponds to in the figure caption to improve the aesthetics of the figure.